

Ruiyu Wang



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Research Interests

Reinforcement Learning & Sequential Decision-Making: Developing and evaluating adaptive decision-making methods for agents operating under uncertainty and changing conditions.

Learning & Planning for Robotics: Studying how learning-based methods can be integrated with planning and feedback for reliable robotic decision-making.

Real-World Robotics & Autonomous Mobility: Interested in integrating perception, decision-making, and learning for autonomous systems operating in complex real-world environments.

Education

University of California, Los Angeles

Los Angeles, CA

M.S. in Computer Science

09/2026 – 06/2028 (*Expected*)

Emory University

Atlanta, GA

Cumulative GPA: 3.93/4.00

08/2022 – 05/2026

Bachelor of Science in Computer Science

Bachelor of Science in Applied Mathematics and Statistics

Selected Research & Engineering Experience

MagicLab Robotics

Suzhou, China

Research Intern

Learning and Memory for Robotic Manipulation

07/2026 – Present

- Conducting model training and evaluation experiments for VLA-based robotic manipulation policies.
- Investigating long-horizon memory mechanisms for retaining task-relevant information across robot interactions.
- Collecting and processing robot demonstration data for policy training and evaluation.

Department of Mathematics, Emory University

Atlanta, GA

Undergraduate Research Assistant

Supervisor: Dr. Yuanzhe Xi

Reinforcement Learning for Multigrid Tuning (HYPRE)

09/2025 – Present

- Formulated multigrid tuning in HYPRE as an RL problem across mixed-type BoomerAMG parameters (strength thresholds, smoothing parameters, coarsening and relaxation types).
- Implemented a policy-based tuner that adapts multi-parameter configurations across solves.
- Benchmarked contextual bandits vs. episodic RL against GPTune/GPTuneBand to assess efficiency and stability.

Emory Graph Mining Lab, Department of Computer Science, Emory University

Atlanta, GA

Undergraduate Research Assistant

PI: Dr. Carl J. Yang

AceSearcher: Improving Reasoning and Search for LLMs via Reinforced Self-Play

03/2025 – 05/2025

- Proposed ACESearcher, a self-play RAG framework where one LLM alternates as Decomposer (breaking multi-hop queries) and Solver (integrating retrieved evidence) to improve reasoning.
- Co-authored the project (4th author), accepted to NeurIPS 2025.

Enhanced Atrial Fibrillation Prediction with Pre-training and Transfer Learning

08/2024 – 02/2025

- Developed hypergraph-transformer pretraining pipelines on a small stroke cohort to learn transferable embeddings.
- Transferred 32-D embeddings to an ESUS cohort (n=510) and integrated clinical features, improving AF risk prediction with lightweight classifiers, yielding 5–15% AUROC gains over from-scratch baselines. Co-authored the project (3rd author); manuscript accepted to AMIA 2026.

Graph-Augmented Sequential Prediction with Large Language Models

05/2025 – 12/2025

- Developed a KG-guided prediction framework on MIMIC-III, integrating structured reasoning chains into LLaMA-8B fine-tuning for interpretable disease detection with 1000 patient visits as training data.
- Achieved AUC/AUPR comparable to traditional ML baselines.
- Led the project end-to-end and authored the study; accepted to AMIA 2026.

Publications & Presentations

- Xu, R., Zhuang, Y., Dong, Z., **Wang, R.**, Yu, Y., Ho, J. C., Zhang, L., Wang, H., Shi, W., & Yang, C. (2025). “AceSearcher: Bootstrapping Reasoning and Search for LLMs via Reinforced Self-Play.”  *NeurIPS 2025* (Spotlight).
- Wang, R.**, Vinh, T., Xu, R., Zhou, Y., Lu, J., Pasquel, F., Ali, M., & Yang, C. (2026). “Knowledge Graph Augmented Large Language Models for Disease Prediction.”  *AMIA 2026 Amplify: Informatics Summit*
- Xie, Y., Wu, Y., **Wang, R.**, Nahab, F., Hu, X., & Yang, C. (2026). “Enhanced Atrial Fibrillation Prediction in ESUS Patients with Hypergraph-based Pre-training.”  *AMIA 2026 Amplify: Informatics Summit*
- Xu, R., Shi, W., **Wang, R.**, Zhou, J., & Yang, C. (2025). “MedAssist: LLM-Empowered Medical Assistant for Scrutinizing and Understanding Electronic Health Records.”  *WWW '25 Companion*, 2931–2934.
- Wang, R.**, Xie, Y., Hu, X., Yang, J. C., & Lu, J. (2026). “BioMedJImpact: A Dataset and LLM Pipeline for AI Engagement and Scientific Impact Analysis of Biomedical Journals.”  *KDD Undergraduate Consortium 2026*.

Industry Experience

- JINGDONG (JD), China** May 2025 – June 2025
AI Intern
- Fine-tuned LoRA models to generate high-fidelity images of JD’s mascot, improving brand-aligned output quality.
- Tongcheng Holdings Limited, China** June 2025 – August 2025
AI Engineer Intern
- Refined the speaker diarization system by fine-tuning the backbone and QMF, reducing EER from 0.87% to 0.37% and improving internal recall from 79% to 96%.
 - Contributed to DeepTrip, an LLM-based travel assistant, designing an evaluation framework using structured LLM prompts to assess factual accuracy, constraint adherence, and response clarity.

Leadership & Service Experience

- Emory University Mathematics Association (EUMMA)** Fall 2025 – Spring 2026
Vice President
- Co-organized academic programming including Honor Research Panel, Graduate School Panel, and Major/Minor Panel; coordinated faculty and student panelists and managed event logistics.
 - Provided competition advising for MCM and Kaggle teams, including team formation, modeling strategy, evaluation methods, and report writing.
 - Led and organized preparation workshops and mock judging sessions to strengthen competitive performance.

Honors & Awards

- Honorable Mention, The Mathematical Contest in Modeling Spring 2025
Dean’s List, Emory College of Arts and Sciences Fall 2023 – Fall 2024
Best Team and Insight Award, DataFest 2024, Emory College of Arts and Sciences Spring 2024

Skills & Interests

- Programming & Systems: Python, C, Java, Linux, Git, Docker, ARM Assembly
- Machine Learning: PyTorch, deep learning, reinforcement learning, model training and evaluation, LLM fine-tuning
- Robotics: ROS 2 (basic), kinematics (basic), motion planning (basic), feedback control (basic)
- Embedded Systems: STM32 (HAL), UART, timers, PWM, interrupts, ADC, PWM-based motor/servo control
- Other Interests: Guitar, music composition, badminton

Relevant Coursework

- Fall 2026: MECH&AE C263A – Kinematics of Robotic Systems; EC ENGR M240A – Linear Dynamic Systems
- Machine Learning: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Probabilistic Machine Learning
- Mathematics & Statistics: Linear Algebra, Mathematical Statistics, Numerical Analysis